

Rickbotics and Mortchanical Mayhem



Version 3
This set of rules is FINAL

Gameplay & Scoring, General:

Each full round of play is two minutes long, and comprises an initial 30s autonomous period that automatically rolls into the 90s “regular” RC period. Players may not touch their robots during game play.

No part of the robot can be moving prior to game play.

There are several scoring opportunities in this game:

- 1. Vat of Acid: Collect balls of acid and get them in your clone tubes**
- 2. Gooble Box: Generate power for the car by pressing the gooble box buttons**
- 3. Mind Blowers: Remove the memories from the holders**
- 4. Car Doors and Hood: Open the doors to lighten the load and check the battery**
- 5. The Car Multiplier: Lift the car for a multiplier**
- 6. The Elevator: Travel up the elevator to a special private spot**
- 7. Portals: Stop the portal for extra points**
- 8. Pickle Rick! Release Pickle Rick (and other characters) down the zip line**

Players must remain on their respective sides of the game table, except in the vat of acid which is a common area accessible to both players.

Course staff are empowered to make all scoring judgments and interpretation of rules. Remember, display unqualified kindness and respect in the face of fierce competition.

Vat of Acid

Description: Remove as much acid as you can and fill your clone tubes for points.

First Tube	0.5 point per ball
Second Tube	1.0 points per ball
Third Tube	1.5 points per ball
Fourth Tube	2.0 points per ball
Fifth Tube	2.5 points per ball
Sixth Tube	3.0 points per ball

Notes:

1. Acid color is irrelevant. It looks great, but makes no difference to scoring.



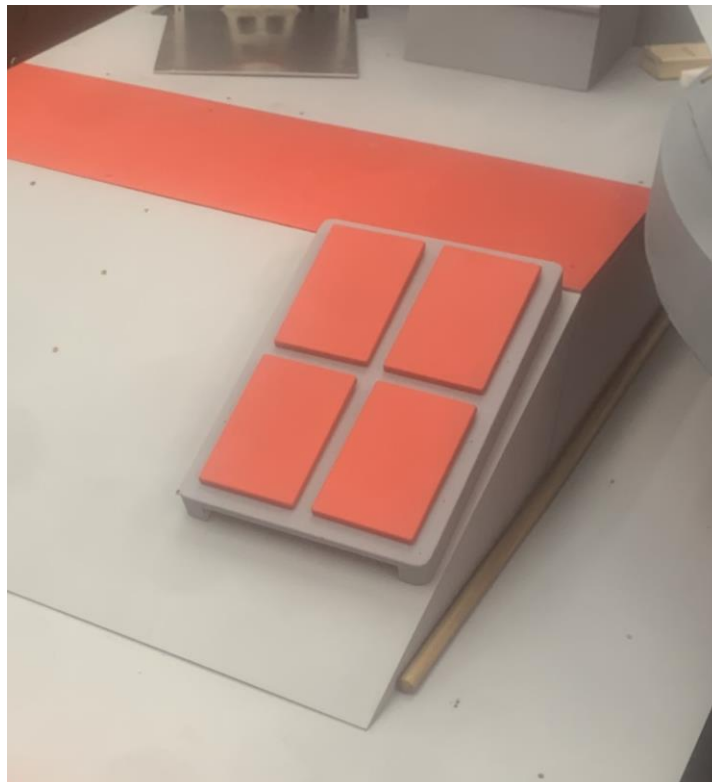
Gooble Boxes

Description: Points are awarded by pressing the buttons of the gooble boxes. Each gooble box has four buttons – an upper pair and a lower pair. Points are awarded based on alternatively pressing each button of a pair.

Upper gooble box buttons	1.0 point per press
Lower gooble box buttons	0.5 point per press

Notes:

1. Consecutive points are awarded by alternating button presses. For instance, at the upper buttons, a right-left-right-left-right sequence would generate 5 points.
2. There is an (electronic) 500 ms second delay imposed between consecutive button presses. Therefore, there is no advantage to press faster than 120 Hz.



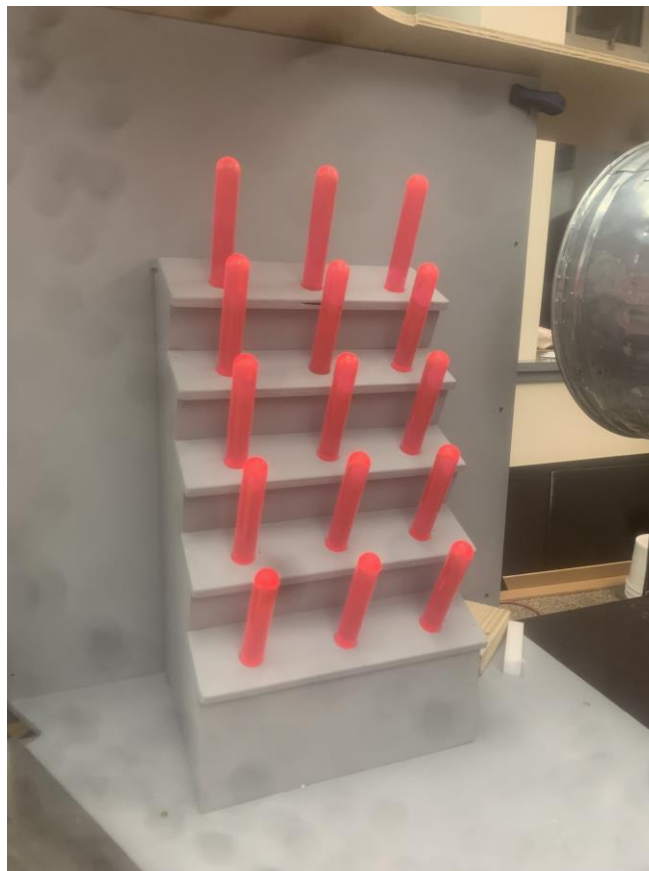
Mind Blowers

Description: Points are awarded by removing mind blowers from their storage shelves. Points are distributed as follows:

Lowest shelf (1)	7 points per memory
Shelf 2	15 points per memory
Shelf 3	23 points per memory
Shelf 4	35 points per memory
Upper Shelf (5)	41 points per memory

Notes:

1. Memories must be completely removed from their holders and cannot rest or touch any part of the shelves.



Car Doors

Description: Open the side doors or front hood for points. Points are awarded based on the maximum angle achieved. Be careful, as the whole car may move on you!

Scoring:

Side Door Angle Zone 1	15 points
Side Door Angle Zone 2	17 points
Side Door Angle Zone 3	21 points

Front Hood Angle Zone 1	31 points
Front Hood Angle Zone 2	35 points
Front Hood Angle Zone 3	40 points

Notes:

DRAFT NOTE: Angles will be marked, the three zones roughly equally spaced from barely open to fully open.



Car Multiplier

Scoring:

The entire car may be tilted to obtain a multiplier for your points. The multiplier is determined by the maximum lift angle obtained. The highest multiplier value is applied to all points, and the multiplier may be obtained at any time during game play. The maximum multiplier obtained by each player will be applied once, and only once. Official referees, course instructors, and staff will determine which multiplier value is obtained.

Multipliers:

Angle 1: 1.2x

Angle 2: 1.5x

Angle 3: 1.8x

Angle 4: 2.0x



SIMILAR TO THE DOORS, the angle zones will be indicated by a gauge or electronic display, and will be roughly equally distributed from zero tilt through when the thrusters bottom-out on the other side (roughly 30 degrees total tilt).

Notes:

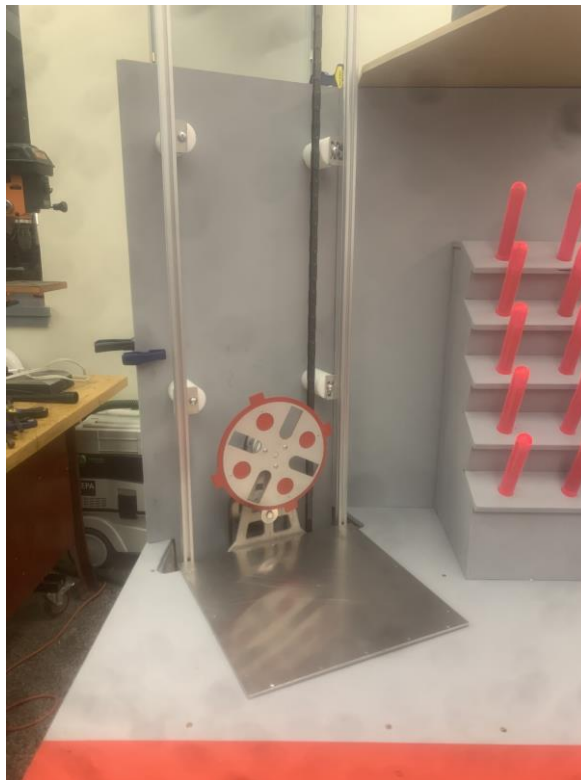
1. The car is a pendulum and can swing to your opponent's side. Such swinging will not generate a multiplier for the opposing player. Judges will determine the player(s) responsible for generating the lift (angle).

2. Players may not wedge any object or otherwise impede their opponent's ability to use the multiplier, except by virtue of actively attempting to tilt the car themselves. Two players simultaneously, actively lifting the car is permitted. Placing a wedge, block, or "inactive" robot as a blocking strategy is in opposition to our principle of gracious professionalism, and not permitted. Don't be "that person" - ugh.

The Elevator

Description: Crank yourself up the elevator to the upper level. Points are cumulative and awarded based on the maximum height achieved. The portals are activated if the elevator reaches the upper shelf.

Zone
First zone: 21 points
Second zone: 39 points
Third zone: 57 points
Fourth zone: 79 points



Notes:

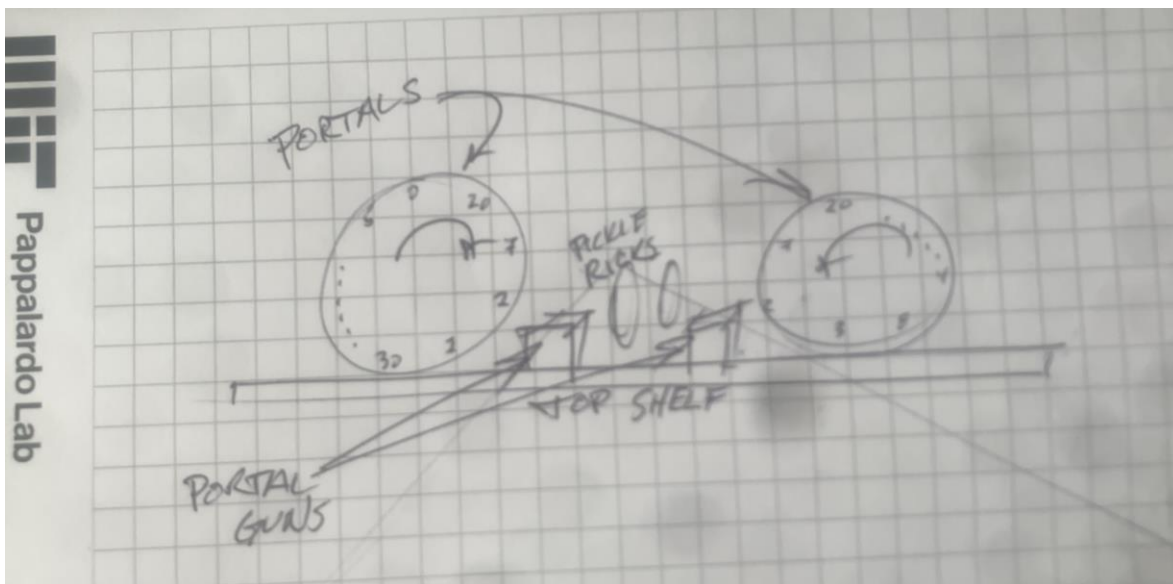
1. Points are awarded once and only once. Traveling up and down repeatedly will not score multiple times.
2. If a height is achieved during autonomous play, it is scored based on the cumulative zones reached during autonomous play. If certain height is obtained during regular play, the score is based on the cumulative zones reached during regular play. A single zone will not be scored for both play periods. For example, an elevator that reaches the first zone during autonomous play

will be scored 42 points. Remaining at this elevation during the RC period does not count for additional RC points. However, should a player continue to crank up during the RC period to the second mark, an additional 39 points will be awarded, for a total of 81 points.

Portals

Description: Portals are activated when the elevator reaches the top shelf. They will continue rotating until the portal gun is pressed. Portal points are awarded based on the digit indicated on the portal when the portal gun is pressed.

Scoring: 0-35 points, based on where the portal stops.



Notes:

1. The portal gun interface is essentially the same as the gooble boxes. That is, it has a flapper button of similar style and press force.

Pickle Rick

Description: Get Pickle Rick down the zip-line for 50 bonus points.

Notes:

1. Pickle Rick needs to slide down the line and be within the footprint of the bottom-most starting zone.



Autonomous Period:

To encourage exploration and learning of electronics in 2.007, the Autonomous Period is 30 seconds. All points are doubled during the autonomous period, except the vat of acid points. If neither competitor wishes to use the autonomous period, they can indicate to the referee and the contest will begin with the 90 second remote control phase.

Scoring Examples:

Game 1:

Blue starts on the platform and autonomously grabs two mind blowers from the bottom shelf. The clock ticks over to RC gameplay, and Blue then drives to the elevator, cranks themselves up to the top lever (activating the portal), pressed the portal gun which stops the portal at the 7 point mark. They try to release Pickle Rick, but run out of time.

Blue's score: $28 [\text{autonomous mind blowers}] + (21+39+57+79) [\text{top level}] + 7 [\text{portal}] = 231 \text{ points}$

Red scooped three balls of acid into the fifth tube, and 2 into the sixth tube during RC play. They opened the car's hood to the second angle zone, and also tilted the car to its maximum angle for a 2x multiplier.

Red's score: $(13.5[\text{acid}] + 35 [\text{hood}]) * 2 = 97 \text{ points}$

Blue prevails.

Game 2:

Blue lifted the car to the 1.5 angle, then quickly drove up the ramp to the mind blowers, removing a mind blower from each of lower four levels (with a very clever mechanism). They started cranking up the elevator, and reached the second level before the game was over.

Blue's score: $(7+15+23+35 [\text{mind blowers}] + (21+39) [\text{Elevator, partly up}]) * 1.5 = 210 \text{ points}$

Red started at the platform level, wrestled with blue to lift the car, and achieved the 1.2x multiplier before they punted and continued to other challenges. They then activated a clever device that automatically pressed the upper gooble box buttons for the remainder of the game, the remaining time happened to be 65s. Their second device cranked up to the top shelf and released Pickle Rick!

Red's score: $(58 [\text{upper buttons}] + (21+39+57+79) [\text{elevator}] + 50 [\text{Pickle Rick}]) * 1.2 = 295 \text{ points}$

What an exciting round!! Red has the higher score and therefore wins.

Rules & Regulations

1. General Principles

- a. These rules are intended to create opportunities to learn engineering.
- b. Those things not specifically forbidden are allowed.

2. Timing

- a. Each round of the contest is 120 seconds long.
- b. For the first 30 seconds, no control signals may be sent to the robot. So that your robot can sense the start of the round, players will trigger the start of the autonomous round with their own light source – this may be a laser pointer, flashlight, phone, etc. Players may not touch the robot to initiate the autonomous period.
- c. If both players agree to not use the first 30 seconds, then the match will last 90 seconds.
- d. Robots may not be touched by players during game play.

3. Winning & Advancing

- a. There will be a single elimination tournament with 32 slots in the bracket. The winner of the tournament is the winner of the contest.
- b. In the tournament, the highest scoring robot in each match advances to the next round. In general, only one machine may advance. If there is a tie, it will be broken by weighing the two systems. The lighter machine prevails.
- c. There are two ways to earn a spot in the tournament bracket – 16 spots are given to the 16 highest players in a ladder, 16 more spots are awarded through a qualification tournament.
- d. The qualification tournament has only one round. Of those who win their match, the top 16 scorers advance into the single elimination tournament. The qualification tournament is run the night before the final tournament. Placement within this tournament bracket is determined by seeding scores. Seeding rounds will be held during the last week of lab time. In the seeding rounds, machines will run on the table unopposed.
- e. Ladder rounds can be held at any point on or after February 24, at any time the lab is open. A student establishes their position on the ladder by: 1) demonstrating to a section instructor, TA, or UA that they can score at least one point, and 2) placing their name on the list according to the points they receive. If a student ties with a score already on the ladder, the student is in the position below that score. The ladder is simply a list of students, ordered according to the points they can obtain during the semester. The ladder may include all students, but only the top 16 scorers secure their position for the final round of play, i.e. the second night of our festivities. The ladder will be closed end-of-day, Friday before Impound Week.

4. Control

- a. When possible, contestants must participate in controlling their own machines. Rarely, schedule conflicts may preclude a player from operating their machine, in which case, with their instructor's permission, another player can be designated.
- b. All control must be accomplished without contacting the robot.
- c. Control may be achieved via radios and, in addition, any another wireless device approved by their section instructor (e.g. TV remote control, laser pointer, recorder, accordion, bagpipes, cell phone, Xbee, Bluetooth, etc).
- d. A contestant may use their single lab issued six channel radio controller. However, contestants may add more degrees of control by using additional controllers such as a TV remote control, laser pointer, receiver, or playstation controller.
- e. Outside Controllers are permitted, but only to supplant the existing 6-channel controllers.
- f. A contestant may not deliberately interfere with the radio or other remote control of the opposing player.
- g. "Assistant drivers" may be used, though only one standard lab RC may be used for the player.
- h. Contestants must wear safety glasses when in the vicinity of the table. Some prescription glasses are acceptable.

5. Robot Configuration

- a. Your entire robot must fit in a *Starting Volume* at the time of impounding and at the beginning of each match when set up on the table. The starting volume is 12" square and 16" tall. The box is "virtual" and has no ceiling or door. These dimensions are smaller than those of the lockers, so you should be able to fit the robot and also your tools and spare materials in your locker. This virtual starting box may be placed anywhere within the colored starting zone at the front of the tables (adjacent to the vat of acid), or within the colored starting zone on the platforms in front of the Mind Blowers and Elevator.
- b. Your entire robot must be made from lab supplied materials and components, fasteners and items in the stock shelves, approved batteries, and electronic components authorized by your section instructor. The tools in the tool kit are considered to be part of your robot kit and can therefore be part of your machine as long as safety and protection of the table are adequately considered in your design. Items that primarily serve to make robots look cool generally will be allowed including LEDs, seven-segment displays, and such items. There are no chemicals in the kit or supply cabinet aside from epoxy, cyanoacrylate adhesive, grease, and such things. For example, baking soda and vinegar would make a great balloon inflation means, but they are not in the kit and cannot be in your robot. Electric motors (or other actuators) taken from appliances (or other household or industrial equipment) are not allowed.
- c. Your "kit" includes up to 6 cubic inches of 3D printed parts. You are also limited to consuming up to 4 cubic inches of support material.
- d. You may use any combination of four lab supplied gearhead motors AND any combination of four lab-supplied continuous rotation and position servos.

- e. Machine weight is limited to 12lbs. You should generally still attend to the weight of your machine as excess weight may lead to poor performance. Also, weight of your system will be used as a tie breaker.
- f. Energy may be stored in batteries, compressed air, elastic strain, and gravitational potential energy. Total stored energy may not exceed 50kJ. This limit will be enforced by the section instructors based on calculations in the lab notebooks. Compressed air may not exceed 50 psi gauge pressure and all air containers can add up to 2 liters maximum. Mechanisms using large amounts of rubber or springs must have adequate safety locks to reduce the chance of accidents. These locks may be removed once all contestants and onlookers are safely away from the device. Safety of all forms of energy storage will be at the discretion of the section instructors and judges.
- g. No boost converters are allowed. Voltage is limited to the 7.4V LiPO batteries. If 7.4 V is to be supplied to any electronics, it must be approved and checked by an instructor AND must be fused. The LiPo batteries are considered 800 mA-hr devices, regardless of their actual markings.
- h. The rechargeable AA -four packs that are typically used with the lab radio receivers are considered 500 mA-hr devices, regardless of the actual labeling on the batteries.
- i. Contestants will be responsible for charging their own batteries, compressed air containers, springs, rubber bands, etc.
- j. Your machine may be reconfigured between rounds. One reason to reconfigure is to accommodate the differences between the right and left sides of the field. You will know at least 5 minutes before you compete which side your machine will be assigned.
- k. You will have access to set up your machine(s) within the starting box prior to each round. You should be able to complete your set-up fully in fewer than three minutes.

6. Sporting Conduct & Safety - Gracious Professionalism

- a. The Vat of Acid is considered common space and useable by both players.
- b. Crossing to the other player's side of the table is not permitted, except within the Vat of Acid.
- c. Damaging the contest table and or control equipment is strictly forbidden.
- d. In the case of destruction deemed by the judges to be accidental, the judges may permit repairs and a rematch.
- e. Contestants and spectators (i.e. any human beings) may not directly affect the motion of the machines or anything else on the table.
- f. It is permissible for a robot to reach outside the boundaries of the table during a round.
- g. Nets or entanglement devices are not permitted, but other defensive devices generally are permitted.
- h. NO DANGEROUS MACHINES. THE JUDGES' DECISIONS ON SAFETY MUST BE RESPECTED AND OBEYED PROMPTLY.

General questions may be asked of UA's, TA's, and Instructors; however, all officially sanctioned rule interpretations will be made by Council of Ricks.